

TaxaTools

Foundation package for the TaxaID ecosystem. Provides taxonomic name handling, LLM provider functions, and shared utilities used by all downstream TaxaID packages.

One of the most difficult aspects of biodiversity science is standardizing taxonomic information. The taxize package was long the standard tool for this, but it is no longer maintained. TaxaTools replaces and extends that functionality: it cleans, verifies, and standardizes taxonomic names across multiple backbones (GBIF, NCBI, WoRMS, Catalogue of Life). Nomenclature inconsistencies between data sources can cause false negatives when priors and likelihoods fail to join on mismatched names. TaxaTools also provides a unified interface for calling large language models (LLMs) from R, with support for Anthropic Claude, Google Gemini, OpenAI, and local Ollama models. LLM-assisted functions can draft Methods and Results text for scientific papers describing how a user arrived at a particular consensus.

Features

- **Name cleaning** – strip authors, subspecies, formatting artifacts (`clean_taxon_names()`)
- **Name verification** – check spelling and resolve against GBIF, NCBI, WoRMS, or Catalogue of Life (`verify_taxon_names()`)
- **Common name lookup** – LLM-assisted conversion of common names to scientific names with optional backbone verification (`common_to_scientific()`); useful when classifier output lacks scientific names
- **Backbone translation** – move names between taxonomic backbones (`change_backbone()`)
- **Column standardization** – rename columns to DarwinCore conventions (`rename_cols()`)
- **LLM providers** – unified interface to Anthropic Claude, Google Gemini, OpenAI, Azure OpenAI (DOI network/VPN required), and local Ollama models
- **Text generation** – LLM-assisted drafting of Methods and Results sections (`draft_methods_text()`, `draft_results_text()`)
- **Report assembly** – combine per-package report sections into a unified document (`assemble_report()`)
- **GBIF census** – enumerate described species per genus from the GBIF backbone (`census_genus_species()`)
- **FAIRe export** – convert TaxaID output tables to FAIRe eDNA checklist format (`to_faire()`)

Installation

```
# From local source (recommended during development)
devtools::install("path/to/TaxaTools")

# Or from the parent TaxaID directory
devtools::install("TaxaTools")
```

Quick Start

```
library(TaxaTools)

# Clean messy names
clean_taxon_names(c(
  "Fundulus parvipinnis (Girard, 1854)",
  "Atherinops affinis",
  "sp. nov."
```

```

))
#> [1] "Fundulus parvipinnis" "Atherinops affinis" NA

# Verify against NCBI
verified <- verify_taxon_names(
  c("Fundulus parvipinnis", "Atherinops affinis"),
  backbone_id = 4
)

# Derive canonical taxon label from taxonomy columns
df <- data.frame(
  family = "Fundulidae",
  genus = "Fundulus",
  species = "Fundulus parvipinnis"
)
create_taxon_names(df, rank_system = c("family", "genus", "species"))

# Call an LLM (provider auto-detected from API keys on library load)
answer <- call_api("What habitat does Fundulus parvipinnis prefer?")

# See how many tokens were used
attr(answer, "tokens")
#> $input
#> [1] 12
#> $output
#> [1] 87

# Print token counts inline (useful during prompt development)
answer <- call_api("Summarise the diet of Atherinops affinis.",
  show_tokens = TRUE)
#> Tokens used - input: 14, output: 63

# Guard against accidentally sending a very large prompt
answer <- call_api(big_prompt, max_input_tokens = 5000)

```

Taxonomic Backbone IDs

Several TaxaTools functions accept a **backbone_id** integer to specify which taxonomic backbone to query or translate to. The supported backbones are:

ID	Backbone
1	Catalogue of Life
3	ITIS
4	NCBI
9	WoRMS
11	GBIF

Full list: <https://verifier.globalnames.org/>

API Keys

Several TaxaID functions require API keys. See the API Setup vignette for where to get keys and how to configure them in `~/Renvirom`.

Key	Purpose	Free?
ENTREZ_KEY	NCBI taxonomy queries	Yes
ANTHROPIC_API_KEY	Claude LLM (default provider)	Paid
GBIF_USER / GBIF_PWD / GBIF_EMAIL	GBIF occurrence downloads	Yes
GEMINI_API_KEY	Google Gemini (optional)	Free tier
AZURE_OPENAI_API_KEY	Azure OpenAI — DOI employees only; requires connection to a DOI computer system or DOI VPN	DOI only

Vignettes

- Cleaning and Verifying Taxon Names – core taxonomy workflow
- Setting Up API Keys – ecosystem-wide API configuration guide

FAIRe Compatibility (eDNA standard terminology)

TaxaID outputs are compatible with the **taxaRaw** and **taxaFinal** classes of the FAIRe eDNA metadata checklist (Takahashi et al. 2025, doi:[10.1002/edn3.70100](doi:%5B10.1002/edn3.70100){.uri}).

Use `to_faure()` to convert any TaxaID match, likelihood, or posterior data frame to FAIRe-compatible column names before export or submission:

```
faure_df <- to_faure(posterior_df, table_type = "taxaFinal")
```

Field mapping (TaxaID → FAIRe):

TaxaID column	FAIRe field	FAIRe class
observation_id	seq_id	taxaRaw / taxaFinal
taxon_name	scientificName	taxaRaw / taxaFinal
taxon_name_rank	taxonRank	taxaRaw / taxaFinal
score	percent_match	taxaRaw / taxaFinal
coverage	percent_query_cover	taxaRaw / taxaFinal
accession	accession_id	taxaRaw / taxaFinal
testid	assay_name	taxaRaw / taxaFinal
kingdom ... species (constructed)	unchanged (Darwin Core)	taxaRaw / taxaFinal
(constructed)	verbatimIdentification	taxaRaw / taxaFinal
(added)	specificEpithet	taxaRaw / taxaFinal
	checkls_ver	projectMetadata

`to_faure()` covers the bioinformatics assignment tables only. Sample-level metadata (**samp_name**, **eventDate**, **decimallatitude**, **env_broad_scale**, etc.) must be joined from field records before submission.

Part of TaxaID

TaxaTools is the foundation package in the TaxaID ecosystem for probabilistic taxonomic assignment. All other packages depend on it for name handling and LLM access.

Ecosystem: TaxaTools -> TaxaFetch -> TaxaHabitat -> TaxaExpect -> TaxaAssign / TaxaMatch -> TaxaLikely -> TaxaAssign -> TaxaFlag

See the TaxaID README for ecosystem overview and installation instructions.

Citation

Lafferty, K.D., 2026, TaxaID – A modular R ecosystem for Bayesian taxonomic assignment: U.S. Geological Survey software release, <https://doi.org/10.5066/xxxxxx>.

Software Requirements

- R ($\geq 4.1.0$)
- An API key for at least one LLM provider (Anthropic, Google Gemini, OpenAI, or Ollama) is needed for text generation functions
- rgbif (for GBIF backbone census; in Suggests)

All dependencies are declared in the DESCRIPTION file and installed automatically.

Developed with Claude Code (Anthropic).